# RAHUL MITRA

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## SUMMARY

Data and AI engineer building optimisation, LLM, and data systems for manufacturing, with experience at Abbott, an STMicroelectronics warehouse AI capstone, Amazon Robotics computer vision research, and Churp, a civic platform built with People's Association and PA ICT (GovTech). Final-year NUS Industrial and Systems Engineering and Computer Science student.

## EDUCATION

**National University of Singapore** **Aug 2023 – Jul 2027**
**Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)**

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Data Structures and Algorithms, Artificial Intelligence, 3D Computer Vision, Operations Research, Simulation, Statistics, Linear Algebra.

## WORK EXPERIENCE

**Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone)** **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine ingesting each site's shelf coordinates, dimensions, and operating rules, letting one model adapt to any STMicroelectronics warehouse floor plan without per-site rework.
- Designed a 4-stage shelf-ranking pipeline: movement profiling from pick frequency, dwell time, and demand; filtering on packaging zone, fit, capacity, and restricted areas; Golden Region priority for fast movers; and logged rationale per decision.
- Defined a historical-replay evaluation comparing 3 metrics against current allocation (travel distance per order, handling time, slow-mover Golden Region occupancy), piloting on 2 AMK warehouses: dice and finished goods.

**Global Data Engineer (Operational AI Systems), Part-Time, Abbott Laboratories (Contract)** **Jun 2026 – Present**

- Developed ADAM, an offline tool-calling assistant on a locally hosted Gemma 4 E4B model (llama.cpp) answering spray-dryer setpoint, process, and signal questions for 6 dryers across 4 global sites, giving engineers self-serve answers to decisions previously requiring APC specialists, with rollout to Abbott servers scheduled.
- Converted 4 sites' Advanced Process Control master files from Excel into 7 steady-state gain models solved as HiGHS linear programs and SLSQP tracking QPs, returning setpoints, binding constraints, and shadow prices in milliseconds across 5 modes: APC replica, single-setpoint push, CV targeting, what-if limit reports, and operator what-ifs.
- Implemented minimum lexicographic constraint relaxation with one LP per candidate, replacing a reference simulator's bisection search, and fixed 2 defects in its quadratic program: a tolerance-band penalty applied to measured rather than predicted values, and an out-of-bounds start point.
- Validated replica outputs against live APC setpoints, matching move direction in 83% of cases and temperature setpoints within 0.12 °C over 24 readings, after tracing a tag-mapping bug pairing one dryer with a second dryer's Seeq tags.
- Qualified each operating point from read-only Seeq pulls via plant-state classification and 30-minute steadiness checks before any solve, and kept LLM output free of computed numbers through JSON-schema-constrained tool calls and automatic checks of each narration against solver results.
- Built a Seeq signal-analytics toolkit (Kalman smoothing, Welch spectra, lag and coherence, PELT change points, matrix-profile motifs and discords, an 18-feature fingerprint) and a 5-stage offline handbook RAG pipeline: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain retrieval.
- Proved production modules identical to research notebooks with exact-equality parity tests, alongside 20 engine tests, HTTP contract tests, a headless notebook runner, and a deterministic model stand-in for GPU-free CI.

**Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract)** **Jan 2026 – Jun 2026**

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Deployed a team-built APC simulator to production for live internal users on Azure App Service, with Docker packaging, access restrictions, and runtime and session hardening.
- Trained regional engineering teams across Asia and Europe on repeatable AI workflows spanning 7 areas: prompting, model selection, benchmarks, cost awareness, agents, validation, and safeguards.
- Strengthened Excel, VBA, Python, master-data, and scheduling workflows with 3 safeguards: validation rules, correction tables, and reproducible intermediate stages, leaving every transformation auditable.
- Documented deployment, access provisioning, and technical handover across 3 procedures, so internal manufacturing and engineer-training users stayed supported after transfer to owning teams.

**Software Engineer (Full Stack & Cloud), People's Association (Pro bono)** **Sep 2025 – Sep 2026**

- Recruited into PA's Sparks Citizens Developer network as a community innovator in recognition of prior public-good initiatives, then designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.
- Owned most software development and deployment across admin and public Next.js frontends, a NestJS backend, and GIS mapping.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Provisioned 10 AWS services with Terraform (ECS Fargate, Multi-AZ RDS Postgres with PostGIS, ALB behind WAFv2, Route 53, ACM, KMS, Secrets Manager, and a Rust auth gateway on Lambda and API Gateway), shipped via keyless GitHub Actions CI/CD on IAM OIDC across 5 repos with gated migrations, RDS snapshots, and auto-rollback.
- Led deployment planning, multi-repo coordination, and stakeholder management in an 11-person team across People's Association, Tengah community partners, and PA ICT (GovTech).

- Surveyed garden sites into geospatial datasets for mapping, plot allocation, and site planning, and designed rules-based balloting across 4 control points (eligibility, conflict resolution, exceptions, admin oversight), keeping allocation auditable.

## RESEARCH EXPERIENCE

**Research Intern, Digital Supply Chain Group (Incoming), A*STAR** **Jan 2027 – May 2027**

- Selected for a 4-month research attachment in ARTC's Digital Supply Chain Group, running alongside final-semester studies.

**Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge)** **Jul 2026 – Oct 2026**

- Tackled an Amazon Robotics brief spanning 3 workstreams (camera ISP, super-resolution, image restoration) to make low-cost cameras rival far pricier sensors, mentored by Dan Shao (CTO, Zelig; former Principal Computer Vision Engineer, iRobot).
- Reproduced a published UNet super-resolution model in PyTorch from broken author code, recovering its undocumented evaluation protocol and training 12 models across 3 scales, 2 datasets, and 2 losses, matching reported results at 8x and landing within 0.78 dB at 2x and 4x.
- Diagnosed a robustness gap where fixed-blur models lost most PSNR gains under unseen blur, then added randomised-blur training and fine-tuning to recover up to 0.7 dB across 3 real-world blur types: lens, out-of-focus, and resampling.
- Authored and presented a super-resolution literature review spanning multi-image, video, and single-image methods, organised around camera information loss and recovery through calibration, extra frames, or learned priors, comparing 4 model families on fidelity and hallucination risk.
- Deployed 3 trained models as a public web app: a one-click Flask inference API on a Colab T4 GPU loading weights from a GitHub release, a secure tunnel on a reserved domain, and a hosted front end for image upload, scale selection, and download.

## PROJECTS

- Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants, Sep 2026): Directed a 5-person team building decentralised drone-swarm coordination in a deterministic Unreal Engine 5.8 C++ simulation; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, versus 2.1 of 8 on average for an operator-run fleet.
- Personal Project, Hailo Edge AI Training and Deployment (Mar 2026): Trained YOLO26 through transfer learning and reshaped its architecture with the Hailo Dataflow Compiler, running document detection on a Hailo-8L rated 13 TOPS at 1.5 W to speed up large-batch scanning.
- Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams, Jan 2026): Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.
- Hackathon Project, Waaah Comics: 24-Hour Gesture-to-Comic Build (Jan 2026): Turned MediaPipe 3D hand landmarks into generated comic strips in a 24-hour sprint with Gemini 2.0 Flash, Veo 3, and Konva.js canvas tooling.
- Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon (Jun 2025): Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.
- Competitor, DSTA BrainHack 2025 CTF (Feb 2025): Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, ranking top 25% of 400+ teams.

## CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

**Independent Bug Bounty Researcher, YesWeHack** **May 2024 – Present**

- Assessed GovTech GBBP12/13, Land Transport Authority, and 13+ other programs for SSRF, CSRF, injection, and access-control flaws, automating recon and disclosure evidence with Burp Suite and Python fuzzing scripts.

**Elderly Care Volunteer, NTUC Health Active Ageing Centre** **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

## NATIONAL SERVICE

**Republic of Singapore Navy, Naval Chef & Base Support Assistant** **Jan 2021 – Jan 2023**

- Compressed a month-long manual analysis of 6 months of food-waste data into a 3-minute Excel VBA reporting workflow inside a security-constrained environment.

## SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, C++, SQL, Bash, Excel VBA
Web and Cloud: React, Next.js, Three.js, NestJS, FastAPI, Flask, Docker, AWS (ECS Fargate, RDS, ALB, WAFv2, Lambda, API Gateway, Route 53, ACM, KMS, Secrets Manager, IAM OIDC), Terraform, AWS SAM, Azure App Service, GitHub Actions and Bitbucket CI/CD, Supabase, Singpass Login and Myinfo (OIDC), API design, GIS mapping
AI and Machine Learning: RAG, AI agents, schema-constrained tool calling, local LLM serving (llama.cpp, Gemma), vector search, LangChain, PyTorch, ONNX, Chroma, BERT/transformer fine-tuning, model evaluation, deep RL (PPO, A2C, DDPG, DQN), GPU training (CUDA) on ASPIRE 2A HPC, Seeq time-series analysis (Kalman filtering, spectral analysis, change-point detection, matrix profile)
Simulation and Operations Research: SimPy, OR-Tools CP-SAT, HiGHS linear programming, SLSQP quadratic programming, constraint relaxation, sensitivity analysis (shadow prices), discrete-event simulation, hybrid flow-shop scheduling
Security: OWASP Top 10, Burp Suite, Wireshark, OAuth2/JWT, SSRF/CSRF/injection testing, fuzzing, secure design
Theory: multi-view geometry (camera calibration, epipolar geometry, PnP, structure-from-motion, bundle adjustment), Bayesian networks, queuing theory, statistical inference (hypothesis testing, regression, bootstrap)
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery